

Identities, Transformations & Equation Structure

Integrated Assimilation Book

For a learner who already knows many algebraic rules but does not yet consistently choose the *right form*.

Governing question: What form makes the requested target cheapest?

1. RECONNECT - you know the moves; can you choose the direction?

You probably know:

- $(a+b)^2 = a^2 + 2ab + b^2$;
- $a^2 - b^2 = (a-b)(a+b)$;
- how to solve linear equations;
- how to substitute a value or expression.

The IOQM step is to decide which move is worth making.

First attempt - write only the first useful line

1. Given $x^2 + x = 1$, find $x^4 + 3x$.
2. Evaluate $49^2 - 31^2$.
3. Given $u + v = 10$ and $uv = 21$, find $u^2 + v^2$.
4. You square both sides of an equation. What must happen before any candidate is accepted?

Do not fully solve unless the target demands it.

The intended first ideas are:

- use the given relation as a rewriting rule;
- factor a difference of squares;
- reconstruct the symmetric target;
- verify candidates in the original equation.

2. DISCOVER - factor and expand are opposite representation choices

Consider $49^2 - 31^2$.

Long calculation treats the two squares separately. Structure sees

$$49^2 - 31^2 = (49 - 31)(49 + 31) = 18 \cdot 80 = 1440.$$

The identity was already known. The new habit is:

Use the direction that resembles the requested information.

Contrast A - expand

You need the coefficient of x in $(x+5)(x-2)$.

Expanding is useful:

$$(x+5)(x-2) = x^2 + 3x - 10.$$

Contrast B - factor

You need the zeros of $x^2 - 25$.

Factoring is useful:

$$x^2 - 25 = (x-5)(x+5).$$

Same algebra. Different target. No direction is automatically "simpler."

Try

For each, choose expand, factor, or leave as is.

1. coefficient of x in $(2x-1)(x+7)$;
2. solve $x^2 - 9 = 0$;
3. evaluate $1003^2 - 997^2$;
4. find the product of the roots of $(x-4)(x+6) = 0$ without expanding.

The point is the decision, not speed.

3. MAKE SENSE - identity vs relation on solutions

An identity is true for every permissible input:

$$(x+1)^2 = x^2 + 2x + 1.$$

A relation on solutions is true only for values satisfying a condition:

$$x^2+x=1.$$

For those admissible values, the relation can be rewritten as

$$x^2=1-x.$$

That is a local rewriting rule. It is not an identity in all real x .

Example - reduce the target, not the variable

Given $x^2+x=1$, find x^4+3x .

From $x^2=1-x$,

$$x^4=(1-x)^2=1-2x+x^2=1-2x+(1-x)=2-3x.$$

Therefore

$$x^4+3x = 2.$$

We never needed the individual values of x .

Why this matters

A common reflex is:

1. solve the quadratic;
2. substitute each root;
3. simplify.

That route may be valid, but it solves more than the target asks. Relation reduction keeps the information in its cheapest form.

Boundary with later polynomial methods

Here we learn the elementary habit "a relation can rewrite higher powers." A later polynomial chapter develops polynomial remainder/reduction as a general method.

4. DISCOVER - substitution must compress structure

Suppose

$$(x+1/x)^2 - 5(x+1/x) + 6 = 0, \text{ with } x \neq 0.$$

The block $x+1/x$ repeats. Set

$$t = x+1/x.$$

The equation becomes

$$t^2-5t+6=0.$$

This is a useful substitution because one symbol replaces a repeated structure and the original restriction $x \neq 0$ remains visible.

Bad substitution test

If a new variable:

- appears only once,
- creates another equation of equal complexity,
- or hides a restriction you later forget,

then it may be renaming rather than simplifying.

Contrast

- $x^4+6x^2+5=0 \rightarrow t=x^2$ compresses the even-power structure.
- $x^2+3x+7=0 \rightarrow t=x+1$ does not automatically make the target smaller.

Ask: what complexity disappeared?

5. MAKE SENSE - symmetric reconstruction

Suppose you know

$$u+v=S \text{ and } uv=P.$$

Then

$$u^2+v^2=(u+v)^2-2uv=S^2-2P$$

and

$$u^3+v^3=(u+v)^3-3uv(u+v)=S^3-3PS.$$

If the target is unchanged when u and v are swapped, test whether sum/product data already determine it.

Example

Given $u+v=10$, $uv=21$,

$$u^2+v^2=100-42=58.$$

Solving u and v individually is unnecessary.

Contrast - when individual values matter

If the target is $u-v$ rather than $(u-v)^2$, order/sign information matters. Sum and product may not decide the requested signed value by themselves.

Boundary with root-coefficient methods

This chapter owns reconstruction from already supplied sum/product information. A later polynomial chapter derives root sum/product from polynomial coefficients.

6. MAKE SENSE - equivalence is stricter than a legal-looking line

A transformation is equivalent if it preserves exactly the same solution set under the recorded conditions.

Reversible examples

- add the same expression to both sides;
- subtract the same expression;
- multiply/divide by a known nonzero constant.

Conditional examples

Squaring

$$A=B \Rightarrow A^2=B^2.$$

The reverse need not hold because $A^2=B^2$ allows $A=B$ or $A=-B$.

So squaring often creates candidates, not automatically solutions.

Dividing by an expression

From $x(x-4)=0$, dividing by x loses the valid branch $x=0$.

Before cancellation, ask whether the factor can vanish.

Clearing denominators

Before multiplying an equation by $(x-2)(x+5)$, record $x \neq 2, -5$.

A later algebraic candidate that violates an original denominator restriction is rejected.

Adopt: "algebraically legal" and "logically equivalent" are not synonyms.

7. TRY - support fades toward independence

Each item is attempted before using the stated support.

Full support available

Given $a+b=12$, $ab=20$, find a^2+b^2 .

Use:

$$a^2+b^2=(a+b)^2-2ab.$$

Answer: 104.

Structural prompt available

Given $x^2=3x-1$, express x^5 as $Ax+B$.

Representation cue: treat $x^2=3x-1$ as a rewriting rule and reduce after every multiplication.

Check after your attempt: $x^5=55x-21$.

Recognition prompt only

Given $p+q=9$, $pq=14$, find p^3+q^3 .

Clue: the target is unchanged when p, q are swapped.

Independent

Given $t^2=2t+3$, reduce t^6-5t^4 to $At+B$.

No method label. Decide the representation, execute, and check.

8. DIAGNOSE - four tempting wrong starts

Error 1: "Expanding is progress"

Why tempting: school exercises often ask for expanded form.

Repair: ask what information the target needs.

Error 2: "If I can solve the variable, I should"

Why tempting: equations invite root-finding.

Repair: test whether the requested expression is already fixed by a relation or symmetry.

Error 3: "A substitution is always helpful"

Why tempting: a new letter looks shorter.

Repair: demand a measurable reduction in repeated structure or degree.

Error 4: "Squaring/cancelling gives an equivalent equation"

Why tempting: the algebraic line is syntactically valid.

Repair: track direction of implication, excluded values and original-equation checks.

9. ADOPT - the target-first checklist

Before writing algebra:

1. What exactly is requested?
2. Which visible form already resembles the target?
3. Would factoring expose a product/zero structure?
4. Would expanding expose coefficients or additive structure?
5. Does a composite block repeat enough to name?
6. Is the target symmetric, so individual variables may be unnecessary?
7. Can a given relation rewrite high powers?
8. Will the next step preserve equivalence?
9. Am I solving more than the problem asks?
10. How will I verify the result independently?

Compressed:

TARGET -> STRUCTURE -> REPRESENTATION -> FIRST MOVE -> CONDITIONS -> CHECK.

10. Historical anchor traces

Use exact historical wording from the validated papers when practicing the full PYQs.

- IOQM-2025-Q01: translate the given percentage statement and notice that the requested quantity is algebraically the same quantity; verified answer 40.
- IOQM-2025-Q21: model the consecutive/integer structure first; a low-degree relation and admissibility condition collapse the search; verified answer 49.
- IOQM-2024-Q05: set ratio variables whose product is 1; one expansion reconstructs the requested difference; verified answer 01.
- IOQM-2024-Q11: reciprocal substitutions make the two supplied constraints comparable; the transformed variables are forced equal and the target is reconstructed; verified answer 12.

These four items are evidence for mechanisms, not a claim of official topic weightage.

11. TRANSFER - the same habit, different surfaces

To inequalities

Before applying a bound, a later inequalities chapter asks which representation exposes the quantity to be bounded. The transformation habit comes from here; bound selection, equality and attainment are taught there.

To polynomials

A later polynomial chapter turns relation rewriting into a canonical polynomial-reduction method and turns symmetric root data into root-coefficient identities. Retrieve the target-first habit; do not duplicate those derivations here.

To functional equations

A later functional-equations chapter uses substitution strategically: choose inputs that expose a relation instead of plugging random values.

To radicals and logarithms

A later radicals-and-logarithms chapter owns its domain and reversibility doctrine. The transferable habit here is to ask whether each transformation preserves equivalence.

To integer equations

A later integer-equations chapter often uses factorisation or substitution to create a finite integer structure, then applies number-theoretic restrictions.

Final belief

I do not manipulate first and interpret later. I read the target, choose the cheapest representation, preserve conditions, and stop as soon as the requested information is determined.

ALG-01 - First-Step Reference

Recognition atlas

Visible structure	First question
difference of squares	factor?
repeated composite block	substitute?
symmetric target in two variables	rewrite via sum/product?
high power + low-degree relation	reduce powers?
radicals/denominators then squaring/clearing	what domain/equivalence check?
target already close to a known identity	reconstruct rather than solve?

Quick router

text

TARGET first

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+--> product/zeros -> factor
+--> coefficient form -> expand
+--> repeated block -> substitute
+--> symmetric target -> use invariants
+--> high power + low-degree relation -> reduce
+--> non-reversible step -> record/check candidates

First-step cards

Factor vs expand

Choose the direction that makes the requested target shorter.

Substitution

Name only a block that genuinely repeats.

Symmetry

Swap the variables mentally. If the target is unchanged, individual values may be unnecessary.

Relation reduction

Write the highest available power as a lower-degree expression and reuse it.

Equivalence

After squaring/dividing/clearing denominators, verify candidates in the original condition.

Contrast strip

- identity vs solution-only relation;
- solving variables vs evaluating a symmetric expression;
- reversible step vs one-way implication;
- elementary relation rewriting here vs general root-coefficient and polynomial-reduction methods in a later polynomial chapter.

ALG-01 - Recognition and First-Line Lab

Rule: write only the first useful mathematical line. Do not complete the solution unless needed to verify that your first line is legal.

1. Evaluate $103^2 - 97^2$.
2. Find the coefficient of x in $(2x-3)(x+8)$.
3. An equation contains four occurrences of $x+1/x$, with $x \neq 0$.
4. Given $a+b=7$, $ab=10$, find a^2+b^2 .
5. Given $x^2=4x-2$, reduce x^5 .
6. Solve $(x+3)^2=(x-1)^2$: what equivalent factorization should be written before solving?
7. Solve $(x-2)(x+5)=0$: should you expand first?
8. Given $u-v=3$ and $uv=10$, find $(u+v)^2$ without solving u,v .
9. In $(x-1)/(x+2)=3$, what restriction must be written before clearing the denominator?
10. Given $t^2+t+1=0$, reduce t^3 .
11. You want p^3+q^3 and know $p+q$ and pq .
12. In $x(x-6)=0$, someone divides both sides by x . What branch must be protected first?
13. $x^4+6x^2+5=0$ contains only even powers. Write a compressing substitution.
14. For $(x+4)^2$ you need a polynomial in powers of x . What direction is useful?
15. For $x^2-16=0$ you need the solutions. What direction is useful?
16. Given $m+n=11$, $mn=24$, target $m-n$. State why sum/product reconstruction alone does not determine the signed target without an order convention.
17. Given $y^2=5y-1$, target y^6+7y . Write the first relation-reduction line.
18. An equation is transformed by an implication-only step and produces candidates 2 and -3. Write the final verification action.

After the first-line attempt, classify each decision using one of:

FACTOR, EXPAND, SUBSTITUTE, SYMMETRY, RELATION_REWRITE, CONDITION_CHECK, STOP/INSUFFICIENT.

ALG-01 - Practice and Transfer Bank

Attempt every item before consulting support or the teacher key.

F0 - reconnect

1. Factor x^2-49 .
2. Expand $(2x-3)^2$.
3. If $a+b=11$ and $ab=24$, find a^2+b^2 .
4. Evaluate 52^2-48^2 structurally.
5. State whether $(x+1)^2=x^2+2x+1$ is an identity or a solution-only relation.

F1 - direct representation choice

6. Evaluate 1004^2-996^2 without long multiplication.
7. If $x^2=5x-3$, reduce x^4 to $Ax+B$.
8. Set a substitution to reduce $(y+1/y)^2-7(y+1/y)+10=0$, $y!=0$.
9. Given $u+v=8$, $uv=12$, find u^3+v^3 .
10. Solve $(x-7)(x+2)=0$ using the form already given.

F2 - standard

11. If $t^2=3t+1$, find t^6 in the form $At+B$.
12. Solve $(2x+3)^2=x^2$ by rewriting as a difference of squares. State why the factorization is equivalent to the original equation.
13. Given $p-q=5$, $pq=14$, find $(p+q)^2$.
14. If $z+1/z=3$, find z^4+1/z^4 without solving for z .
15. Explain why cancelling $(x-1)$ from $(x-1)(x+4)=0$ is not equivalent unless the $x=1$ branch is retained.

F3 - disguised

16. A number x satisfies $x^2-x-1=0$. Compute x^8 in the form $Ax+B$.
17. If $a+b+c=0$, simplify $a^3+b^3+c^3$ structurally.
18. Given $r+s=5$ and $r^2+s^2=13$, find r^4+s^4 without solving r,s .
19. A real nonzero x satisfies $x+1/x=4$. Find x^5+1/x^5 .
20. Solve $(x+6)^2-x^2=20$ using a structure-preserving rewrite. Identify why each transformation is reversible.

F4 - preliminary-style selection

21. Given $q^2=2q+1$, compute q^7-13q^3 as $Aq+B$.
22. If $a+b=6$ and $ab=5$, determine $(a-b)^4$ without finding a,b .
23. Let $w+1/w=2$, $w!=0$. Find w^8+1/w^8 using structure, not individual root solving.
24. The equation $(x^2+3x)^2-7(x^2+3x)+10=0$ is given. Introduce one substitution and solve for the *possible values of the repeated block only*.
25. Given $x^2+x=1$, evaluate x^4+3x without solving for x .

Transfer

26. Representation change: rewrite a high-power target under $x^2=4x-1$ so that it becomes linear in x ; explain why this is a bridge to polynomial methods without using polynomial-remainder theory here.
27. Context change: a rectangle has side lengths u,v with known perimeter and area. Explain which symmetric expressions can be found without reconstructing each side.
28. Functional bridge: if a functional equation repeatedly contains $f(x)+f(1-x)$, propose a strategic substitution/combination goal without developing functional-equation theory here.
29. Integer bridge: transform $mn+m+n=35$ into a factored form suitable for integer case analysis.
30. Cross-domain: explain how the question "what form makes the target cheapest?" should be reused before choosing an inequality representation.
31. Construct one case where expansion is cheaper than factorisation and justify the target.
32. Construct one case where factorisation is cheaper than expansion and justify the target.

ALG-01 - Independent Mixed Mastery Check

No default hints. No method labels. For each problem, decide the first move yourself.

1. Evaluate $1007^2 - 993^2$.
2. If $a+b=13$ and $ab=36$, find a^3+b^3 .
3. If $x^2=2x+5$, express x^5 as $Ax+B$.
4. Given $p-q=4$ and $pq=21$, find p^2+q^2 .
5. Solve $(3x+4)^2=(x+2)^2$ by an equivalent factorization. State why the factorization does not create extra candidates.
6. Let $y+1/y=5$, $y \neq 0$. Find y^3+1/y^3 without solving for y .
7. If $u^2+u+1=0$, determine u^{2026} .
8. An equation contains (x^2+3x) repeatedly. Write a substitution that lowers structural complexity and state the test for whether it is useful.
9. Explain precisely why dividing $x(x-4)=0$ by x is not an equivalent transformation.
10. You need the coefficient of x^2 in $(x+1)(x+2)(x+3)$. State the cheapest first representation move.
11. You need the zeros of $x^2-10x+21$. State the cheapest first representation move and solve.
12. Given $m+n=9$, $mn=14$, decide whether the signed value $m-n$ is determined without an order convention. Justify.
13. Given $t^2=4t-1$, compute t^4-15t as $At+B$.
14. Solve $(x+1)/(x-3)=2$. Record every excluded value before transforming and verify the final candidate.
15. WHY NOT: a student solves r,s separately from $r+s=8$, $rs=15$ before finding r^2+s^2 . Explain why the route is oversized and give the cheaper first line.
16. Transfer: an integer problem has $ab+a+b=48$. Transform it to a factor equation without enumerating a,b yet.

For items 1, 3, 5, 11 and 15, also state the clue that determined your first move.